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An artificial neural network for exploring the relationship between learning activities and students' performance

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ABSTRACT

This paper identifies the most significant learning factors impacting undergraduate academic performance using artificial neural networks (ANNs) and controlled student data collection. As higher education becomes increasingly common and important, finding the best ways to help students optimize their studies is vital. Questionnaires gathered data on student behaviours and achievement from five classes within a semester, constraining variability to compare learning activities directly. The questionnaire captured engagement, psychological factors, effort, course load, time management, and performance data. Statistical and exploratory analysis investigated the dataset. A multilayer perceptron model was developed, using backpropagation and cross-validation to optimize predictive accuracy. The model identified class attendance, sleep quality, and questioning during lectures as most correlated with high grades. Additional patterns emerged around research participation, motivation, cramming, and theoretical studying. This research demonstrates new techniques for associating detailed study behaviours with academic achievement through strictly controlled student data collection and the application of artificial neural networks for predictive modelling. The constrained variability in the dataset allows for isolating the impact from specific learning activities. The controlled student data and machine learning-driven predictive modelling provide information on the optimal grouping of student effort across engagement, health, and studying factors.

1. Introduction

Higher education is more common than before. As most higher education teaching approaches rely on the self-discipline of the students, a lecturer normally has less monitoring on the students. Therefore it is important for the students and lecturers to identify the learning activities that are able to contribute to the understanding of the students in the subject [1]. This in turn can lead to better educational outcomes and contribute to the success of students in their future careers.

In today's fast-paced world, students and lecturers are growing increasingly disconnected. Lecturers have different expectations for their students and are left disappointed when students fail to conform to them [2]. Likewise, students also find themselves in a disarray with their expectations of the type of education they are to receive at university [3]. Furthermore, students find themselves in an increasing amount of stress for getting better grades and that trend does not seem to be going anywhere anytime soon [4] In fact, if anything, it is growing. Students now face more stress not only due to external stressors like social media [5], but also academic pressure in performing better in a highly competitive environment that is created around the students [4]. On the other hand, we are now in an era where new information is constantly being added to their subject matter. With so much information available, it is easy to lose focus. It is therefore important to know not only where time is more efficiently spent studying for a subject but also to identify the factors that have a causation effect on a student's performance.

In an ideal situation, teachers track their students progress to enhance the efficiency of the learning process continuously throughout a semester with the knowledge of where and what to fix and what factors to watch out for. This further extends to the teachers being able to have a way of knowing if the topic alongside with their teaching methods is effective and results in higher productivity for the students. However, most of the time the only point of reference that the teachers can evaluate if their teaching methods are sufficient is the results of their students in the various coursework and exams that they take along with any separate feedback they receive from their students [6]. Teachers are also unable to know how engaged the students are with their teaching and the subject at hand. At the same time, artificial intelligence (AI) use is increasing in the world with increased applications of domains under it such as data mining, prediction etc. Similarly, progress is also seen in the world of education with Chen et al. [7] doing a review on this field and finding 4519 publications from 2000 to 2019. They found the use of AI in a wide field of subdomains notably performance prediction. Similarly [8] who also did

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a review found the term "predict" being a part of one of the main topics in the use of AI with similar results also being mentioned by Holmes et al. [9]. These show not only the growth of AI in the field and its increasing focus in the world of education [10], but also the viability of using it for prediction of factors and aspects. Garett and Young [11] explored the use of AI in social audio and the potential future it has and in more recent times [12] used ANN in COVID-19 modelling to predict future patients and deaths. As seen in these reviews however, one of the areas where AI use is lacking is in the area of predicting factors and learning related activities that affect student performance. Most papers dealing with prediction do so at predicting student performance or behaviour [7–9,13,14] where it leaves a small gap in expanding the use of AI in education through the use of other sub-domains. To address this issue, this study proposes an artificial neural network (ANN) with primarily collected data to measure the impact of different factors allowing the teachers and educators to know what factors correlate positively or negatively with the student's grades in their class. This can allow them to design more effective teaching strategies and interventions to support struggling students. Further, this study proposes to do so whilst having the data be collected in a much more controlled environment as well allowing for more accurate representative results.

The main aim of this paper is to identify the most significant factors that affect student performance. To do so, it focuses on the design of the questionnaire for the collection of data on student's learning activities and the processing of the collected data to evaluate the factors that affect student's performance. The remainder of the article is divided into six sections. First the considerations during the designing phase of the questionnaire will be discussed. Then the content of the questions being asked will be discussed. The next section will then discuss the factors affecting student's academic performance before the design of the questionnaire for data collection. The collected data will then be analysed using statistical approaches and exploratory approaches. An ANN will then be constructed to provide a predictive model for student's performance in terms of grades using the learning activities as inputs. The hyperparameters of the trained ANN will be analysed to investigate the significance of the inputs, i.e. the learning activities on the output, i.e. the student's performance.

This paper demonstrates new techniques for associating detailed study behaviours with academic achievement through the combination of strictly controlled student data collection and application of artificial neural networks for predictive modelling. The constrained variability in the dataset allows for isolation of the impact from specific learning activities, providing new insights into optimal grouping of student effort across engagement, health, and studying factors.

2. Review of relevant work

The use of AI in education is widespread, one of the most common areas and use cases is in prediction using different models. Dogan et al. [15] did a systematic review examining 276 publications on AI in online distance education. They found use cases in AI in various fields including for teaching/learning, prediction of student behaviours and enabling of personalized learning. An important conclusion they reached was on the importance of using human-centered AI rather than purely technically driven solutions. Rastrollo-Guerrero et al. [16] in a review found 64 papers using various AI and machine learning models in attempting to predict student performance, student dropout, student activities and recommended activities and resources. In their review, they found ANN, Decision Trees (DT) and Support Vector Machines (SVM), models as the most commonly used in these papers.

Looking at each of the models, DT were used in [17] to predict student performance by measuring their cognitive and non-cognitive features. They were also studied in [18] in a review on decision tree algorithms prediction performance.

Moving on, Hussain et al. [19] did a comparative study using Naive Bayesian (NB), ANN, SVM, DT and regression in predicting student difficulties in learning and found ANN and SVM as the most accurate. Likewise Hernández-Blanco et al. [20] found in a systematic review that ANN in deep learning based models achieved higher accuracies than machine learning counterparts in 67% of the papers they reviewed. Musso et al. [21] predicted key educational outcomes such as degree completion and grade point average using a backpropagated ANN. They used various background information from the students to map and reach those outcomes.

Zhao et al. [22] did a study using logistical regression and SVM in predicting student performance to see if they would get certification and managed to achieve an 84% accuracy with SVM. Villagrá-Arnedo et al. [23] did a similar study using SVM to predict student performance and categorize them into 3 categories of performance. Hashemi Petrudi et al. [24] did a study using for performance measurement in higher education institutions using Fuzzy Delphi method and Best Worst Method in identifying a list of performance indicators.

3. Questionnaire design consideration

3.1. Performance factors

When discussing the factors that could affect performance, there are both factors inside the classroom as well as those outside. Table 1 shows the different factors affecting student performance and the studies associated with them. Looking at factors inside the classroom, the significance of student attendance is always a major discussion in every institution. Some have the options of attending or watching pre-recorded online lectures but for those that do not have that option, lack of attendance means missing out the subject material being taught. Caviglia-Harris [25] shows that students who attend class more often end up with better academic performances in comparison to those who do not. Similarly, Ancheta et al. [26] found similar results with student attendance affecting student performance. Academic workload is the next important factor with Kizito et al. [27] and Yang [28] both finding that increased academic workload can lead to decreases in academic performance as well as increases in burnout. Academic workload here is represented through both the number of classes a student has in a semester as well as the amount of work/assignments they have.

Interest in the subject being studied is also an important factor. Interest in a matter often leads to putting more effort and performing better in related tasks overall. Afzal and Ali [29] mentions that students who were engaged enough with a subject to investigate it before the formal teaching were found to more likely score higher marks for that subject. Another similarly related factor is the effort from the student to study the materials before attending the class. This is the case where students study the lecture material or perform the practical tasks before the actual class in order to be more prepared. Chen and Lin [30] found that students who do this on average end up doing better in class and having higher performance scores. Onah and Anamezie [31] also found similar results in secondary school students learning physics where academic interest led to increased performance and achievement.

Part of the interest in classes as well as the effort put into learning can also be described by the student's self motivation. Steinmayr et al. [32] and Kusurkar et al. [33] both found self motivation to be a very important and impactful factor towards student performance with increased motivation having a positive effect on not only performance but on almost all aspects of a student's learning including factors such as student well-being as well as overall engagement and happiness with their studies [34].

There are some aspects that a student does not necessarily have full control of. Stress is one such factor. Saqib and Rehman [35] and Pascoe et al. [36] show that stress has been shown to negatively affect one's life in all aspects and this includes academic performance. Sleep is another important factor that helps regulate the body and the mind. Phillips et al. [37] mentions how lack of quality sleep has led to detrimental effects in memory, focus, and cognitive function all of which are related to studying regardless of in the classroom or self-learning.

Table 1
Factors affecting student performance.

Factors	Student attendance	Interest in class	Stress	Sleep	Self-motivation	Academic workload
Caviglia-Harris [25]	X					
Afzal and Ali [29]		X				
Phillips et al. [37]				X		
Kizito et al. [27]						X
Onah and Anamezie [31]		X				
Saqib and Rehman [35]			X			
Steinmayr et al. [32]					X	
Kusurkar et al. [33]					X	
Ancheta et al. [26]	X					
Yang [28]						X
Pascoe et al. [36]			X		X	
Chen and Lin [30]		X				

3.2. Data gathering

Megel and Heermann [38] discussed multiple ways to perform data gathering such as through interviews, focus groups, document reviews, questionnaire surveys and so on. Megel and Heermann [38] also discussed how questionnaires, since allowing for anonymity, are a good option to use if anonymity is needed. If sensitive data is collected, as it often requires non-anonymous data collection, it will result in detrimental effects to response rates as more will be unwilling to participate. Therefore being able to do the survey anonymously is an encouraged requirement. Furthermore, Navarro-Rivera and Kosmin [39] discuss how a questionnaire allows minimal involvement time by the participants. In a world where attention is measured in milliseconds, a survey shines its light in comparison to other data gathering methods such as interviews or ethnographies for time and attention required. The next reason that a questionnaire is favourable comes from the question of "what data is required?". Megel and Heermann [38] also further discusses how, while interviews allow for high-quality qualitative data, when only quantitative data that no further explanation or in-depth answers are required, questionnaires are better suited. A further advantage is that questionnaires allow for the flexibility of the researchers to choose and base their questionnaire around existing ones. This gives an advantage of having a reference to an established work [40]. This allows easier documentation, as there will not be unplanned spontaneous modification to the flow in the data collection process.

Because questionnaires maintain anonymity, participants have the option to participate in the study without the researchers' knowledge. Therefore in the next subsections we will discuss the non-response bias and response bias before going into some practices that are discouraged in designing a questionnaire.

3.3. Non-response bias

When sampling is performed, not all members of the target group respond, and thus not a full circle of answers can be received but rather only answers from that group are received. This will give rise to non-response bias [41]. Choi and Pak [42] mentioned a big issue with questionnaires is the effect of participation bias in the outcome to make the results inaccurate. This is vital in this study as participation bias potentially happens for students who did not score as highly in the subject. This would sway the results heavily in favour of the top students and thus make the study invalid as it fails to consider all the students of different varieties.

Thus, anonymity is practised to counter this issue. Megel and Heermann [38] discusses how an advantage of questionnaires is that as the respondents do not have to show themselves or reveal any personal details, it gives them the extra comfort from behind an anonymous screen which will allow for more responses. Choi and Pak [42] then talks about how non-monetary incentives such as a personalized message sent through email may help getting more respondents to perform the survey. The questionnaire will also be non-disguised, meaning the participants will be briefed of the use of their data and the outcomes that is to achieve. This gives the survey an extra human touch as the participants will not feel like lab rats undergoing tests. This potentially leads to higher response rates. Toepoel and Schonlau [43] and Kennedy and Ouimet [44] further discussed how giving a questionnaire's introduction and explanation section a more human touch will raise empathy in the participants and thus increase the number of participants.

As the presence for non-response bias can be assumed, the data and program will have to be adjusted accordingly to deal with the issue. If it is found that the participants who answered were all of those who got >60% for example, then the data analysis will be adjusted to see where these students spent their time that allowed them to get these higher scores. Harris and Namboodiri [45] recommends a basic inverse relationship that can be established with the results in that an assumption could be made that students who do not share the same factors like the higher scoring students can be expected to not score as well. This can still help with pattern identification as it would give a base as to what to look out for in a student's performance in a semester.

3.4. Response bias

Biases may still arise when the participants respond to a survey. This is coined as response bias. Due to the nature of the study, most questions asked should be as close-ended as possible, leaving little room for interpretation or subconscious misrepresentation of the questions asked to avoid response bias. Quantitative answers provide another solution for reducing the response bias. Therefore if the nature of the study allows, most of the questions should have quantitative response. Still, the questions must be worded such that they do not cause any sort of response bias. Ornstein [46] recommended that the wording must be concise, valid, and succinct, with phrasing to be done in as much of a neutral demeanour as possible. This also helps in keeping the questions as straight to the point as possible, leaving any unnecessary wordings behind.

3.5. Logistics consideration

Incentives and human touch alone cannot make a good questionnaire that can maximize the response rates and the accuracy of the responses. For the purpose of maximizing the response rate, Arora [47] suggested that the questionnaire should be made simple and straightforward, Ornstein [46] mentioned how the questions should not require assistance in answering them and the questions should flow logically. If the questions are too complex, the response rates will take a hit as the participants may abandon the questionnaire when they have issues understanding the questions [48]. Burchell and Marsh [49] brings up questionnaire length as another important factor in keeping the response rates high. Long questionnaires lead to interest lost in participants. When taking into consideration the target group being university students, time and attention are not plentiful resources. The questionnaire will, therefore, be kept short where possible. A total

completion time of fewer than two minutes would help keeping the interest up while completing the questionnaire.

Most of the algorithms used to predict student performance are either limited by the algorithm itself or by certain contexts of data gathering and analysis. Most of the related literature did not perform primary data gathering, instead they received their information based on other sources such as the school registry. This causes limitations as only certain factors can be extracted this way. Most of these literature only use Grade Point Average (GPA) results with the students background information as factors for consideration [50–52]. Factors that were not considered are ones that involve the engagement and psychological conditions of the students such as the stress levels in the semester or how engaged they were with a subject, which can only be found through directly asking them. This work will perform primary data gathering as this provides the flexibility to collect any relevant data that are considered important for the study. As for the literature that did their own data gathering, it is mostly done at a mass level where large groups of students are generalized as a whole and processed collectively. This approach ignores the effect of the factors such as different subjects, different students of different ages, and different teaching methods. To counter this, a non-probability sampling method was undertaken in the form of the study asking students who were in the same class so that the differentiating factors to be analysed would be from the students themselves, allowing the rest of the variables would be controlled. By controlling the data, it firstly allows for much greater real world accuracy with the data that is collected being more consistent and reliable. As a consequence of that, the results obtained are then more representative too and more importantly, can paint a picture for that specific group of students. Next, the environment be controlled more for data collection also allows for less noise to be present in the data which helps the ANN in training. This is helpful as it allows for a more data-driven environment to be present which in turn helps the ANN in its training. This also enables the comparisons between different classes in order to analyse the factors importance change between classes.

3.6. Questionnaire design

The questionnaire is divided into multiple sections of which the participants are uninformed. The purpose of the sectioning is solely for the researcher to compare and analyse the data. The questionnaire itself, is asked onto students participating in a specific class in a specific semester meaning all education and university related factors are controlled as the participants would be in the same environment. This is true for all 5 modules asked and allows for the study to be unique in its approach where the results obtained would be much more real world accurate and for it to provide better discussion.

The 10 questions chosen to be asked in this study were not backed by an industry expert but rather were supported by the various research works looked at by this study. The most significantly related and impactful factors were then scoured and formed the theoretical model of which the questions stem from, the theoretical sections are discussed below. The other factor on how the 10 questions were defined comes from the anecdotal hypothesis of which variables would be the most applicable when discussing students of this particular university. The importance of this is that by asking more applicable questions, it allows the results obtained in the end to then be more significant and real world accuracy.

This questionnaire is divided into six sections composing the theoretical base which comprise of pre-standing interest, psychological condition, personal effort, semester structure, time management, and lastly the actual grade of the student. The first three sections, pre-standing interest, psychological condition, and personal effort, are factors intrinsic to the students. The following two sections, semester structure and time management, focus on the logistics of the students life. The last section provides the ground truth to which the effect of different factors are analysed. All of these combined help make up the conceptual model of the various factors affecting student performance.

Section A – Students pre-standing interest in the subject

This section focuses on the student's interest in the subject. It aims to capture the relationship of the interest a student has in a topic with their performance and effort in the subject. The questions are answered on a linear scale with values ranging from 0 to 100 with 0 representing "never" and 100 representing "very often".

Question 1: How often did you do research on the subject before it was taught?

This question aims to identify both how much a student knows about a course before it is introduced as well as the amount of interest and effort they are willing to put into it. This is based on the belief that the students who engage enough to investigate a subject before it is being taught are more likely to score higher marks for that subject [29].

Question 2: How often did you ask a question about anything that was taught in the subject?

This question aims to see how engaged a student is with the subject. Asking questions to clarify anything that is taught in the classroom is a good sign that a student pays attention in the classroom and has at least some form of engagement with the lecturer. This question works on the basis that the students who actively try to ask questions on the subject matter and therefore are engaging in self-improvement are more likely to score in the subject. One thing this question failed to account for, however, is the students who do not need to ask anything as they have no issue understanding any part of the content.

Question 3: How often did you study the material to be taught before class?

This question aims to see how dedicated a student is on a subject. It seeks to see if they would take the route of performing the work/studies required before they are even taught of it. This is based on the idea that students who do this on average end up doing better in class and having higher performance scores [30].

The next question was answered with a range from 0 to 100 whereby 0 denotes "not interested" and 100 denotes "very interested".

Question 4: How would you rate your interest in the subject?

This question directly asks the section's headline of their interest to the student. This question seeks to get a direct view of how much the students gauge their interest in the subject. It will be used to investigate the correlation with previous questions.

Section B – Student's external factors in life

This section asked questions regarding a participant's life outside the classroom. It aims to find the link between the factors outside of the classroom and performance inside the classroom. The presumption is that certain negative factors that are unrelated to the classroom can still affect a student's performance. The questions are also answered in a linear scale with values ranging from 0 to 100 with 0 representing "never" and 100 representing "very often".

Question 5: How often would stress from outside affect you in the classroom?

This question aims to see the effects of stress on a student's performance. Stress is a known factor that negatively affects life in all aspects [35]. Likewise, the higher the stress level of a student, the more likely they are to perform badly in a class [35]. This question does not ask students about their stress level directly as everyone perceive their stress levels differently, therefore potentially causing the data to be skewed. Instead, it asked the participants how much their stress